Engineering - Graduate Level Assessment 11 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Find the variance of the random variable, Z = X + Y if X and Y are not independent.
   (a) VarZ = VarX + VarY - 2 E[{X - E(X)}{Y - E(Y)}]
   (b) VarZ = |VarX - VarY|
   (c) VarZ =VarX +VarY
   (d) VarZ = VarX + VarY + Cov(X,Y)
   (e) VarZ = (VarX + VarY) / 2

2. While operating on variable frequency supplies, the AC motor requires variable voltage in order to
   (a) extend the motor's lifespan.
   (b) increase the motor's efficiency.
   (c) reduce the motor's noise.
   (d) maintain a constant speed.
   (e) improve the motor's torque.

3. An SCR has half cycle surge current rating of 3000 A for 50 Hz supply. One cycle surge current will be
   (a) 2121.32 A.
   (b) 6000 A.
   (c) 2000 A.
   (d) 7000 A.
   (e) 5000 A.

4. In an experiment involving the toss of two dice, what is the probability that the sum is 6 or 7?
   (a) 5/36
   (b) 3/12
   (c) 6/36
   (d) 1/6
   (e) 17/36

5. Find the input impedance of a 50 ohm line terminated in + j 50 ohms, for a line length such that $= \pi$ radian.

(a) 100 ohms

(b) 0 ohms

(c) - 50 j ohms

(d) -100 j ohms

(e) + 25 j ohms

True / False (10 marks)

Answer True or False

6. True or False: The radiation resistance of an antenna that radiates 8 kW and draws 20 A is 50 ohms.

7. True or False: In the binary number system, the first digit from right to left is called the Least Significant Bit (LSB).

8. True or False: The pressure drop in a tube with water flowing at 1 fps over 10 ft is 8.22 psf.

9. True or False: The energy dissipated in a 4 resistor with a current of i(t) = sin(t) is E(1) = 1.5 J, E(2) = 3 J, E(3) = 4.5 J.

10. True or False: For an 80% modulation of the audio portion in a TV signal, the carrier swing is 20 kHz and the frequency deviation is 40 kHz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the amount of information conveyed when the sum of a rolled pair of dice is revealed to be seven. Analyze how this relates to concepts of information theory and probability.

12. Discuss the relationship between a time-varying magnetic field, described by B = B_0 $e^{b}ta$_z, and the resulting electric field. Derive the expression for the electric field produced and analyze its implications in engineering applications.

End of Paper